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Pulse duration measuring apparatus

Abstract

A pulse duration measuring apparatus includes a polarizing beam splitter for splitting a pulsed laser beam into a first laser beam and a second laser beam, a first mirror for reflecting the first laser beam traveling toward the polarizing beam splitter, a second mirror for reflecting the second laser beam traveling toward the polarizing beam splitter, a first quarter wavelength plate disposed between the polarizing beam splitter and the first mirror, a second quarter wavelength plate disposed between the polarizing beam splitter and the second mirror, an optical path length changing unit for moving the first mirror or the second mirror to change the length of the respective optical paths, a nonlinear crystal body for allowing a combined laser beam to pass therethrough, and a photodetector for measuring an optical intensity of the combined laser beam that has passed through the nonlinear crystal body.

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to a pulse duration measuring apparatus having a plurality of optical parts for measuring pulse duration of a pulsed laser beam.

Description of the Related Art

(2) Wafers with a plurality of devices such as integrated circuits (ICs) and large-scale integration (LSI) circuits formed in respective areas demarcated on their face side by a plurality of intersecting projected dicing lines are divided along the projected dicing lines into individual device chips by a laser processing apparatus. The device chips will be used in various electric appliances such as mobile phones and personal computers.

(3) The laser processing apparatus includes a chuck table for holding a wafer thereon, a laser beam applying unit for applying a pulsed laser beam to the wafer held on the chuck table, and a moving mechanism for moving the chuck table and the laser beam applying unit relatively to each other in X-axis directions indicated by an arrow X and Y-axis directions indicated by an arrow Y. The laser beam applying unit includes a laser oscillator for emitting the pulsed laser beam, a beam condenser for converging the laser beam emitted by the laser oscillator onto the wafer held on the chuck table, and an optical system disposed between the laser oscillator and the beam condenser (see, for example, JP 2012-002604A).

SUMMARY OF THE INVENTION

(4) In the laser processing apparatus described above, the optical system of the laser beam applying unit is serviced for maintenance periodically or as desired in order to keep a required level of processing quality based on the pulsed laser beam. The various maintenance processes that are performed include an inspecting process for measuring the pulse duration of the pulsed laser beam emitted by the laser oscillator and checking whether the measured pulse duration is a desired pulse duration or not.

(5) The inspecting process for measuring the pulse duration of the pulsed laser beam is carried out by a pulse duration measuring apparatus that is relatively large in size. The pulse duration measuring apparatus operates as follows. First, the pulse duration measuring apparatus is positioned in the vicinity of the laser processing apparatus to be serviced for maintenance. Then, separate beam branching means is used to branch a laser beam from the pulsed laser beam emitted from the laser beam applying unit of the laser processing apparatus. The branched laser beam is guided to the pulse duration measuring apparatus that measures the pulse duration of the branched laser beam. The pulse duration measuring apparatus is tedious and time-consuming to use because it is troublesome to install and adjust the pulse duration measuring apparatus before it is actually able to make measurements.

(6) It is therefore an object of the present invention to provide a pulse duration measuring apparatus that is capable of efficiently measuring a pulse duration without branching a laser beam from a pulsed laser beam out of a laser processing apparatus.

(7) In accordance with an aspect of the present invention, there is provided a pulse duration measuring apparatus having a plurality of optical parts for measuring pulse duration of a pulsed laser beam. The pulse duration measuring apparatus includes a laser beam entry element for receiving a pulsed laser beam, a polarizing beam splitter for splitting the pulsed laser beam from the laser beam entry element into a first laser beam traveling along a first optical path and a second laser beam traveling along a second optical path, a first mirror for reflecting the first laser beam traveling along the first optical path toward the polarizing beam splitter, a second mirror for reflecting the second laser beam traveling along the second optical path toward the polarizing beam splitter, a first quarter wavelength plate disposed between the polarizing beam splitter and the first mirror, a second quarter wavelength plate disposed between the polarizing beam splitter and the second mirror, an optical path length changing unit for moving the first mirror or the second mirror along the optical path of the first laser beam or the second laser beam to change a length of the optical path, a nonlinear crystal body for allowing a combined laser beam to pass therethrough, the combined laser beam including respective returning laser beams of the first and second laser beams

reflected respectively by the first and second mirrors and combined by the polarizing beam splitter, and a photodetector for measuring an optical intensity of the combined laser beam that has passed through the nonlinear crystal body. The laser beam entry element, the polarizing beam splitter, the first mirror, the second mirror, the first quarter wavelength plate, the second quarter wavelength plate, the optical path length changing unit, the nonlinear crystal body, and the photodetector are disposed on a planar surface.

(8) In the pulse duration measuring apparatus according to the present invention, since the components from the laser beam entry element to the photodetector are disposed on a planar surface, the pulse duration measuring apparatus that is of a planar configuration can measure a pulse duration while being placed on a chuck table of a laser processing apparatus. The pulse duration measuring apparatus according to the present invention eliminates the problems of the conventional pulse duration measuring apparatus in that it is tedious and time-consuming to use as the pulse duration of a pulsed laser beam has to be measured by branching a laser beam from the pulsed laser beam emitted from the optical system of the laser beam applying unit.

(9) The above and other objects, features and advantages of the present invention and the manner of realizing them will become more apparent, and the invention itself will best be understood from a study of the following description and an appended claim with reference to the attached drawings showing a preferred embodiment of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of a pulse duration measuring apparatus according to an embodiment of the present invention and a laser processing apparatus;

(2) FIG. 2 is an enlarged perspective view of the pulse duration measuring apparatus illustrated in FIG. 1; and

(3) FIG. 3 is an enlarged perspective view illustrating the manner in which the pulse duration measuring apparatus illustrated in FIG. 2 is used to measure a pulse duration.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

(4) A pulse duration measuring apparatus according to a preferred embodiment of the present invention will be described hereinbelow with reference to the accompanying drawings.

(5) FIG. 1 illustrates in perspective the pulse duration measuring apparatus, denoted by **10**, according to the present embodiment and a laser processing apparatus **1** where the pulse duration of a pulsed laser beam can be measured by the pulse duration measuring apparatus **10**.

(6) As illustrated in FIG. 1, the laser processing apparatus **1** includes a holding unit **3** that is disposed on a base **2** and that holds a plate-shaped workpiece thereon, a laser beam applying unit **6** for applying a pulsed laser beam to the plate-shaped workpiece on the holding unit **3**, alignment means **7** having a camera **7a** for capturing an image of the workpiece on the holding unit **3**, a moving assembly **4** for processing-feeding the laser beam applying unit **6** and the holding unit **3** relatively to each other and moving the alignment means **7** and the holding unit **3** relatively to each other, and a frame **5** including a vertical wall **5a** erected on the base **2** behind the moving assembly **4** and a horizontal wall **5b** extending horizontally from an upper end portion of the vertical wall **5a** in overhanging relation to the holding unit **3**.

(7) The holding unit **3** includes a rectangular X-axis movable plate **31** movably mounted on the base **2** for movement in the X-axis directions, a rectangular Y-axis movable plate **32** movably mounted on the X-axis movable plate **31** for movement in the Y-axis directions that are perpendicular to the X-axis directions, a hollow cylindrical support post **33** fixedly mounted on an upper surface of the Y-axis movable plate **32**, and a rectangular cover plate **34** fixedly mounted on an upper end of the support post **33**. A chuck table **35** is disposed in the cover plate **34** and extends

upwardly through an oblong hole defined in the cover plate **34**. The chuck table **35** is rotatable about its vertical central axis by rotary actuating means, not illustrated, housed in the support post **33**. The chuck table **35** has a circular suction chuck **35a** that is made of an air-permeable porous material and that lies substantially horizontally. The suction chuck **35a** is fluidly connected to suction means, not illustrated, through a fluid channel extending through the support post **33**.

(8) The moving assembly **4** includes an X-axis feeding mechanism **42** and a Y-axis feeding mechanism **44**. The X-axis feeding mechanism **42** converts rotary motion of an electric motor **42a** into linear motion with a ball screw **42b** and transmits the linear motion to the X-axis movable plate **31**, moving the X-axis movable plate **31** in one of the X-axis directions or the other along a pair of guide rails **2a** that are disposed on the base **2** and that extend in the X-axis directions. The Y-axis feeding mechanism **44** converts rotary motion of an electric motor **44a** into linear motion with a ball screw **44b** and transmits the linear motion to the Y-axis movable plate **32**, moving the Y-axis movable plate **32** in one of the Y-axis directions or the other along a pair of guide rails **36** that are disposed on the X-axis movable plate **31** and that extend in the Y-axis directions.

(9) The horizontal wall **5b** of the frame **5** houses therein an optical system, not illustrated, of the laser beam applying unit **6**. The laser beam applying unit **6** includes a beam condenser **6a** disposed on the lower surface of a distal end portion of the horizontal wall **5b**. The camera **7a** of the alignment means **7** is disposed on the lower surface of the distal end portion of the horizontal wall **5b** at a position adjacent to the beam condenser **6a** in one of the X-axis directions. The camera **7a** includes an ordinary image capturing device, such as a charge coupled device (CCD), for capturing images with visible light, infrared ray applying means for applying infrared rays to a workpiece, an optical system for catching infrared rays emitted from the infrared ray applying means, and an image capturing device, such as an infrared CCD, for outputting an electric signal representing the infrared rays caught by the optical system.

(10) The laser processing apparatus **1** includes a control unit, not illustrated. The control unit includes a computer and controls the moving assembly **4**, the laser beam applying unit **6**, and the alignment means **7**, which are electrically connected to the control unit. Display means **8** is disposed on an upper surface of the horizontal wall **5b** and electrically connected to the control unit. The control unit controls the display means **8** to display processing conditions and images captured by the camera **7a** of the alignment means **7**.

(11) The optical system, not illustrated, of the laser beam applying unit **6** includes a laser oscillator for emitting a pulsed laser beam having a predetermined pulse duration. When the laser processing apparatus **1** is serviced for maintenance periodically or as desired, the pulse duration measuring apparatus **10** measures the pulse duration of the pulsed laser beam emitted from the laser oscillator, and checks whether the measured pulse duration is a desired pulse duration or not.

(12) FIG. 2 illustrates in enlarged perspective the pulse duration measuring apparatus **10** illustrated in FIG. 1. As illustrated in FIG. 2, the pulse duration measuring apparatus **10** includes a circular base plate **11** having a flat lower surface. The pulse duration measuring apparatus **10** also includes, on a planar upper surface **11a** of the base plate **11**, a laser beam entry element or a rectangular prism **12** to which a pulsed laser beam is applied vertically from above, a polarizing beam splitter **13**, disposed centrally on the planar upper surface **11a**, for splitting the pulsed laser beam reflected from a reflecting surface **12a** of the laser beam entry element **12** into a first laser beam traveling along a first optical path L1 and a second laser beam traveling along a second optical path L2, with an inner reflecting surface **13a** including a multilayer coating, a first mirror **15A** for reflecting the first laser beam that has traveled along the first optical path L1 from the polarizing beam splitter **13** back toward the polarizing beam splitter **13**, a second mirror **15B** for reflecting the second laser beam that has traveled along the second optical path L2 from the polarizing beam splitter **13** back toward the polarizing beam splitter **13**, a first quarter wavelength plate **14A** disposed between the polarizing beam splitter **13** and the first mirror **15A**, and a second quarter wavelength plate **14B** disposed between the polarizing beam splitter **13** and the second mirror **15B**.

(13) The pulse duration measuring apparatus **10** further includes an optical path length changing unit **16** for moving the first mirror **15A** along the optical path **L1** of the first laser beam to change the length of the optical path **L1**, a nonlinear crystal body **17** for allowing a combined laser beam to pass therethrough, the combined laser beam including respective returning laser beams of the first and second laser beams reflected respectively by the first and second mirrors **15A** and **15B** and combined by the polarizing beam splitter **13**, and a photodetector **18** for measuring the optical intensity of the combined laser beam that has passed through the nonlinear crystal body **17**.

According to the present embodiment, the first optical path **L1** starts from the reflecting surface **13a** of the polarizing beam splitter **13** and returns from the first mirror **15A** back to the reflecting surface **13a** of the polarizing beam splitter **13**, and the second optical path **L2** starts from the reflecting surface **13a** of the polarizing beam splitter **13** and returns from the second mirror **15B** back to the reflecting surface **13a** of the polarizing beam splitter **13**.

(14) The laser beam entry element **12** has the reflecting surface **12a** that is inclined at an angle of 45° to the optical path of the pulsed laser beam applied vertically from above. The reflecting surface **12a** reflects the pulsed laser beam applied vertically from above to travel horizontally, thereby changing the optical path of the pulsed laser beam toward the polarizing beam splitter **13** disposed centrally on the flat upper surface **11a** of the circular base plate **11**.

(15) The optical path length changing unit **16** includes a stepping motor **161**, an externally threaded screw **162** having an end coupled to an output shaft of the stepping motor **161**, and a nut **163** having an internally threaded hole operatively threaded over the externally threaded screw **162**. The nut **163** is integrally coupled with the first mirror **15A**. When the stepping motor **161** is energized, it rotates the externally threaded screw **162** about its central axis, moving the first mirror **15A** in one of the directions indicated by an arrow **R**, i.e., along the first optical path **L1**.

(16) The stepping motor **161** and the photodetector **18** are electrically connected to a control unit **100**. The control unit **100** can store the optical intensity of the combined laser beam, the optical intensity being detected by the photodetector **18**, as an electric signal and can also control the stepping motor **161**. The control unit **100** sends an operation signal to the stepping motor **161** to move the first mirror **15A** accurately along the pulsed first laser beam traveling along the first optical path **L1** in one of the directions indicated by the arrow **R**. The control unit **100** may not necessarily be included in the pulse duration measuring apparatus **10**, but the stepping motor **161** and the photodetector **18** may be electrically connected to the control unit, not illustrated, of the laser processing apparatus **1**, so that the control unit of the laser processing apparatus **1** can be used to control the pulse duration measuring apparatus **10** to measure the pulse duration of the pulsed laser beam.

(17) Operation and advantages of the pulse duration measuring apparatus **10** as applied to the laser processing apparatus **1** will be described below with reference to FIG. 3 as well as FIGS. 1 and 2.

(18) The circular base plate **11** of the pulse duration measuring apparatus **10** is set to dimensions that can be held under suction on the chuck table **35** of the laser processing apparatus **1**. For example, the circular base plate **11** is set to a size commensurate with the suction chuck **35a** or larger dimensions.

(19) For measuring the pulse duration, the pulse duration measuring apparatus **10** is placed on the suction chuck **35a** of the chuck table **35**, and the suction means, not illustrated, is actuated to generate a negative pressure that is transmitted to the suction chuck **35a** to hold the workpiece under suction thereon. Then, the moving assembly **4** is actuated to move the chuck table **35** and hence the pulse duration measuring apparatus **10** to a position directly below the camera **7a** of the alignment means **7**. Then, the camera **7a** is energized to capture an image of the pulse duration measuring apparatus **10** to thereby detect the position of the center of the reflecting surface **12a** of the laser beam entry element **12** disposed on the flat upper surface **11a** of the circular base plate **11**.

(20) After the alignment means **7** has detected the position of the center of the reflecting surface **12a** of the laser beam entry element **12**, the moving assembly **4** is actuated to move the pulse

duration measuring apparatus **10** until the center of the reflecting surface **12a** of the laser beam entry element **12** is positioned directly below the beam condenser **6a** of the laser beam applying unit **6**. Then, the laser beam applying unit **6** is energized to emit a pulsed laser beam LB0 (see FIG. 3) from the beam condenser **6a**. The pulsed laser beam LB0 has a wavelength of 355 nm, for example, and an average output power level low enough not to process or damage the optical components of the pulse duration measuring apparatus **10**.

(21) As illustrated in FIG. 3, the pulsed laser beam LB0 emitted from the beam condenser **6a** is reflected by the reflecting surface **12a** of the laser beam entry element **12** and applied to the polarizing beam splitter **13**. The reflecting surface **13a** of the polarizing beam splitter **13** transmits therethrough the P-polarized light of the pulsed laser beam LB0 from the laser beam entry element **12** as a first laser beam LB1 traveling along the first optical path L1, and reflects the S-polarized light of the pulsed laser beam LB0 as a second laser beam LB2 traveling along the second optical path L2, so that the P-polarized light and the S-polarized light are branched as the first laser beam LB1 and the second laser beam LB2 from the pulsed laser beam LB0.

(22) The P-polarized first laser beam LB1 that has passed through the reflecting surface **13a** of the polarizing beam splitter **13** is converted into a circularly polarized laser beam when it is transmitted through the first quarter wavelength plate **14A** disposed between the polarizing beam splitter **13** and the first mirror **15A**, and is then reflected by the first mirror **15A**. When the first laser beam LB1 is reflected by the first mirror **15A**, the direction of rotation of the circularly polarized laser beam is converted into an opposite direction, and when the first laser beam LB1 passes through the first quarter wavelength plate **14A**, it is converted into an S-polarized laser beam. The S-polarized laser beam is then applied as a returning laser beam LB1' of the first laser beam LB1 to the polarizing beam splitter **13** and reflected by the reflecting surface **13a** thereof to travel toward the photodetector **18**, whereupon the returning laser beam LB1' is applied to the nonlinear crystal body **17**.

(23) The S-polarized second laser beam LB2 that has been reflected by the reflecting surface **13a** of the polarizing beam splitter **13** is converted into a circularly polarized laser beam when it is transmitted through the second quarter wavelength plate **14B** disposed between the polarizing beam splitter **13** and the second mirror **15B**, and is then reflected by the second mirror **15B**. When the second laser beam LB2 is reflected by the second mirror **15B**, the direction of rotation of the circularly polarized laser beam is converted into an opposite direction, and when the second laser beam LB2 passes through the second quarter wavelength plate **14B**, it is converted into a P-polarized laser beam. The P-polarized laser beam is then applied as a returning laser beam LB2' of the second laser beam LB2 to the polarizing beam splitter **13** and reflected by the reflecting surface **13a** thereof to travel toward the photodetector **18**, whereupon the returning laser beam LB2' is applied to the nonlinear crystal body **17**.

(24) As described above, the pulsed laser beam applied to the nonlinear crystal body **17** is a combined laser beam LB3 that includes the returning first laser beam LB1' and the returning second laser beam LB2' that are combined by the polarizing beam splitter **13**. The combined laser beam LB3 includes the P-polarized laser beam and the S-polarized laser beam. According to the present embodiment, the nonlinear crystal body **17** is arranged such that the ratio of transmittances at which the P-polarized laser beam and the S-polarized laser beam of the combined laser beam LB3 are transmitted through the nonlinear crystal body **17** varies depending on the ratio at which the waveforms of the P-polarized laser beam and the S-polarized laser beam overlap each other, thereby changing the optical intensity of a combined laser beam LB4 that is emitted from the nonlinear crystal body **17**.

(25) The control unit **100** according to the present embodiment energizes the stepping motor **161** of the optical path length changing unit **16** to move the first mirror **15A** along the pulsed first laser beam traveling along the first optical path L1, thereby changing the length of the first optical path L1 for the returning laser beam LB1' that is reflected by the first mirror **15A** and that returns to the

polarizing beam splitter **13**, and controls the photodetector **18** to detect a change in the optical intensity of the returning laser beam **LB1'**. The changes in the length of the first optical path **L1** correspond to double the distance that the first mirror **15A** moves therealong.

(26) FIG. **3** illustrates in a lower section a graph representing how the ratio of transmittances at which the P-polarized laser beam and the S-polarized laser beam of the combined laser beam **LB3** are transmitted through the nonlinear crystal body **17** varies, as plotted on the basis of the optical intensity detected by the photodetector **18**. The graph has a vertical axis representing the ratio of transmittances and a horizontal axis representing time, i.e., picoseconds (ps), corresponding to the changes in the length of the first optical path **L1**, obtained by dividing the changes in the length of the first optical path **L1** by the speed of light. The graph also illustrates an approximate curve **S** based on the plotted ratio values.

(27) A point represented by **P0** in the graph is a point where the optical intensity detected by the photodetector **18** is maximum and the waveforms of the P-polarized laser beam and the S-polarized laser beam of the combined laser beam **LB3** do not overlap each other at all. The optical intensity detected at the point **P0** is regarded as a reference (0%) for the ratio of transmittances at which the P-polarized laser beam and the S-polarized laser beam of the combined laser beam **LB3** are transmitted through the nonlinear crystal body **17**. In other words, the optical path length changing unit **16** is operated to make the length of the first optical path **L1** smaller than the length of the second optical path **L2**. As the first mirror **15A** is moved by the optical path length changing unit **16** to increase the length of the optical path, the waveforms of the P-polarized laser beam and the S-polarized laser beam of the combined laser beam **LB3** start overlapping each other. As the ratio of overlapping increases, the optical intensity of the combined laser beam **LB4** emitted from the nonlinear crystal body **17** when the combined laser beam **LB3** passes through the nonlinear crystal body **17** decreases.

(28) A minimum point **P1** of the approximate curve **S** in the graph illustrated in the lower section of FIG. **3** is a point where the waveforms of the P-polarized laser beam and the S-polarized laser beam of the combined laser beam **LB3** fully overlap each other, i.e., a point where the length of the first optical path **L1** and the length of the second optical path **L2** coincide with each other, and the optical intensity of the combined laser beam **LB4** emitted from the nonlinear crystal body **17** when the combined laser beam **LB3** passes through the nonlinear crystal body **17** is lowest. As the first mirror **15A** is moved by the optical path length changing unit **16** to increase the length of the optical path, the ratio of overlapping between the waveforms of the P-polarized laser beam and the S-polarized laser beam of the combined laser beam **LB3** is lowered, and the optical intensity of the combined laser beam **LB4**, the optical intensity being detected by the photodetector **18**, increases. When the waveforms of the P-polarized laser beam and the S-polarized laser beam of the combined laser beam **LB3** finish overlapping each other, the detected optical intensity becomes maximum at a point **P2** where the ratio of transmittances is of the same value as the ratio of transmittances at the point **P0**. A waveform thus derived that is represented by the approximate curve **S** is highly correlated to the pulsed waveform of the pulsed laser beam **LB0**. The pulse duration of the pulsed laser beam **LB0** is calculated from the approximate curve **S** according to the following calculating procedure.

(29) According to the embodiment described above, a change **X1** in the length of the first optical path **L1** from the point **P0** to the point **P2** of the approximate curve **S** is 12000 μm , for example. Dividing 12000 μm by the speed of light (300 μm) per picosecond (ps) results in 40 ps. From the distribution of the ratios of transmittances corresponding to the change **X1**=12000 μm illustrated in the graph in FIG. **3**, a change **X2** in the length of the first optical path **L1** corresponding to the pulse duration, denoted by **W**, is calculated as a change in the optical path length between points **P3** and **P4** at a ratio **Q2** of transmittances that is 1/2 of a ratio **Q1** of transmittances at the minimum point **P1** of the approximate curve **S**. The change **X2** detected according to the present embodiment is 3180 μm , and the pulse duration **W** measured according to the present embodiment is specified

with respect to a value obtained by dividing the change $X2$ by the speed of light, i.e., $300\text{ }\mu\text{m/ps}$, as a reference. More specifically, $3180\text{ }\mu\text{m}\div 300\text{ }\mu\text{m/ps}=10.6\text{ ps}$ is calculated as a reference, and 10.6 ps is divided by $2^{\text{sup.}1/2}$ (≈ 1.41) that represents a parameter on the assumption that the pulsed waveform of the pulsed laser beam **LB0** is a Gaussian waveform. The value ($\approx 7.5\text{ ps}$) thus obtained by the division is determined to be the pulse duration W as a measured value, which is stored in the control unit **100**. It is then determined whether the measured pulse duration W is a desired pulse duration or not. If the measured pulse duration W is not a desired pulse duration, then the laser oscillator that emits the pulsed laser beam **LB0** is adjusted or replaced.

(30) The above process of measuring the pulse duration is by way of example only, and various other processes of calculating pulse durations are known in the art. Therefore, an actual process of determining a pulse duration using the pulse duration measuring apparatus **10** is not limited to the process according to the above embodiment.

(31) According to the above embodiment, the pulse duration measuring apparatus **10** that is of a planar configuration can measure a pulse duration while being placed on the chuck table of the laser processing apparatus. The pulse duration measuring apparatus **10** eliminates the problems of the conventional pulse duration measuring apparatus in that it is tedious and time-consuming to use as the pulse duration of a pulsed laser beam has to be measured by branching a laser beam from the pulsed laser beam emitted from the optical system of the laser beam applying unit.

(32) According to the above embodiment, the optical path length changing unit **16** is provided as means for moving the first mirror **15A** along the optical path. However, it may be provided as means for moving the second mirror **15B** along the optical path. Such a modification also allows the same measurement as described above with reference to FIG. **3** to be made.

(33) The present invention is not limited to the details of the above described preferred embodiment. The scope of the invention is defined by the appended claim and all changes and modifications as fall within the equivalence of the scope of the claim are therefore to be embraced by the invention.

Claims

1. A pulse duration measuring apparatus having a plurality of optical parts for measuring pulse duration of a pulsed laser beam, comprising: a laser beam entry element for receiving a pulsed laser beam; a polarizing beam splitter for splitting the pulsed laser beam from the laser beam entry element into a first laser beam traveling along a first optical path and a second laser beam traveling along a second optical path; a first mirror for reflecting the first laser beam traveling along the first optical path toward the polarizing beam splitter; a second mirror for reflecting the second laser beam traveling along the second optical path toward the polarizing beam splitter; a first quarter wavelength plate disposed between the polarizing beam splitter and the first mirror; a second quarter wavelength plate disposed between the polarizing beam splitter and the second mirror; an optical path length changing unit for moving the first mirror or the second mirror along the optical path of the first laser beam or the second laser beam to change a length of the optical path; a nonlinear crystal body for allowing a combined laser beam to pass therethrough, the combined laser beam including respective returning laser beams of the first and second laser beams reflected respectively by the first and second mirrors and combined by the polarizing beam splitter; and a photodetector for measuring an optical intensity of the combined laser beam that has passed through the nonlinear crystal body, wherein the laser beam entry element, the polarizing beam splitter, the first mirror, the second mirror, the first quarter wavelength plate, the second quarter wavelength plate, the optical path length changing unit, the nonlinear crystal body, and the photodetector are disposed on a planar surface.
